St. George's University
Grenada, West Indies

Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 01

The Back

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 2: pg 86- 101
- Online Textbook: Chapter 2 https://www.clinicalkey.com/student/content/book/3-s2.0-B9780323934237000022

- Pay special attention to the "**In the Clinic**" boxes for the clinical correlations for this topic
    - Blue boxes in online text
    - Green boxes in hard copy

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1076	Describe the structure of the scapula and clavicle, and their function in attachment of the appendicular (shoulder) to the axial skeleton (trunk).
SOM.MKII.BPM1.1.MSK.1.ANAT.1077	Describe the general difference in function between the superficial (extrinsic) and deep (intrinsic) muscles of the back.
SOM.MKII.BPM1.1.MSK.1.ANAT.1078	Describe the general attachments, innervations, functions and major blood supply of the superficial and deep muscles of the back.
SOM.MKII.BPM1.1.MSK.1.ANAT.1079	Describe the general extent of the thoracolumbar fascia and how it encloses the intrinsic muscles of the back
SOM.MKII.BPM1.1.MSK.1.ANAT.1080	Describe the general attachments, innervations, functions and major blood supply of the erector spinae muscles.
SOM.MKII.BPM1.1.MSK.1.ANAT.1081	Describe the general location, innervations and actions of the transversospinalis muscles.
SOM.MKII.BPM1.1.MSK.1.ANAT.1082	Describe the differences in action between unilateral and bilateral contraction of the muscles of the back.
SOM.MKII.BPM1.1.MSK.1.ANAT.1083	Describe the cutaneous innervation of the back.
SOM.MKII.BPM1.1.MSK.1.ANAT.1084	Describe the location of the lymph nodes into which the lymphatics of the skin of the back drain.
SOM.MKII.BPM1.1.MSK.1.ANAT.1085	List the borders of the triangle of auscultation and define its clinical significance.

Lecture Overview

- Introduction to the MSK System
- Movements of the Back
  - Movements of the Vertebral Column
  - Movements of the Shoulder Girdle (Scapula)
  - Movements of the Shoulder Girdle (Humerus at the Glenohumeral Joint)
- Back muscles
  - Extrinsic Back Muscles
  - Intrinsic Back Muscles
- Thoracolumbar Fascia
- Lymphatic Drainage of Back
- Cutaneous Innervation of Back
- Clinical Anatomy: Triangle of Auscultation, Triangle of Petit
- Surface Anatomy of the Back

Musculoskeletal System

- Muscles move the skeleton.
- Skeleton = Several bones articulating via joints.
- Muscles attach to bones and must cross joints to be able to move them.
- Skeletal muscles are under voluntary control (Somatic efferent).
- Different types of movement (e.g., flexion, extension, adduction, abduction, medial rotation, lateral rotation, ...)

What do you need to know ?

- Be able to identify each muscle on a cadaver or photograph of a cadaver or imaging
- General attachments (bone to bone) unless from an obvious landmark (covered in Imaging)
- Be able to name the nerve(s) that innervates the muscle
- Be able to name the major action(s) for each muscle
- Be able to describe the deficit due to a lesion of the nerve, and/or describe how the patient will present

How are muscles named ?

- Based on various characteristics:

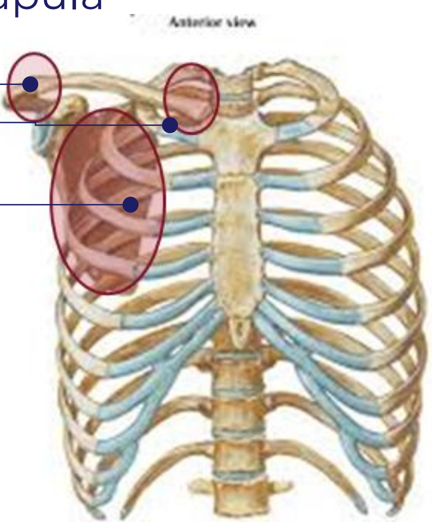
1. **Location** (e.g., **Brachialis**: located in the arm (**brachium**); **Temporalis**: Found near the temporal bone of the skull)
2. **Shape** (e.g., **Deltoid**: Triangular shape, resembling the Greek letter delta (Δ);**Trapezius**: Shaped like a trapezoid).
3. **Size** (e.g., **Gluteus maximus**: largest muscle in the gluteal region, **Adductor brevis**: A smaller adductor muscle compared to others.
4. **Direction of Muscle Fibers** (e.g., **Rectus abdominis**: Fibers run straight up and down (**rectus** = "straight"), **External oblique**: Fibers run diagonally.
5. **Number of Origins (Heads)** (e.g., **Biceps brachii**: Two origins ("bi-" = "two"), **Triceps brachii**: Three origins ("tri-" = "three").
6. **Points of Attachment:** (e.g., **Sternocleidomastoid**: Originates from the **sternum** and **clavicle** and inserts on the **mastoid process**.
7. **Action** (e.g., **Flexor carpi radialis**: Flexes the wrist (carpi), **Extensor digitorum**: Extends the fingers (digits)

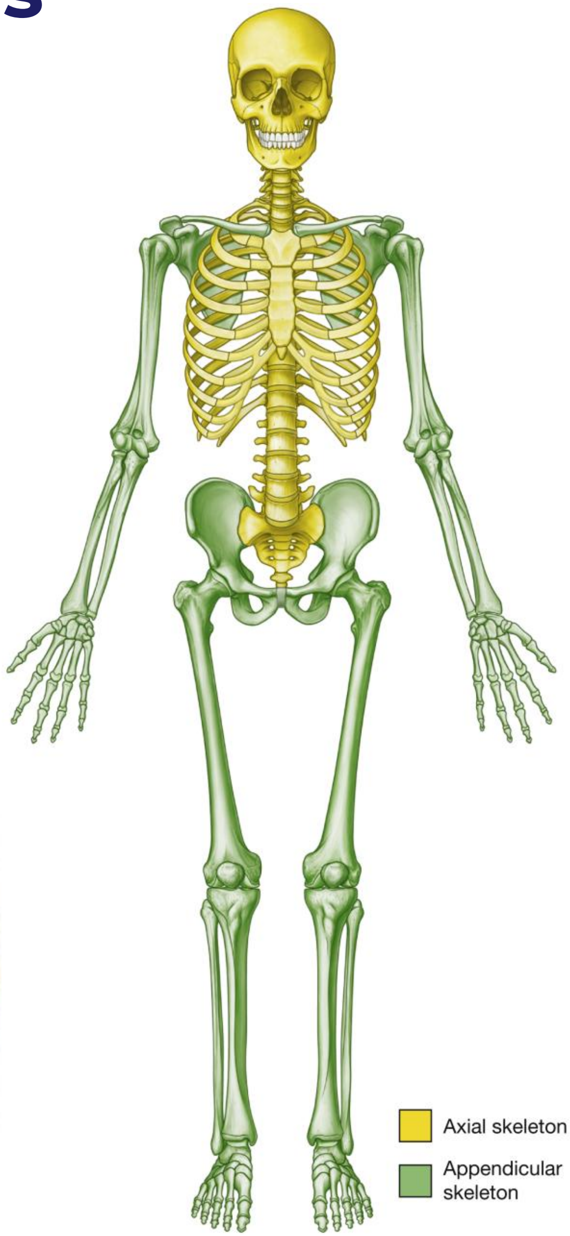
- **Often it is a combination of factors** (e.g., **Flexor digitorum superficialis**: Indicates its action (**flexor**), the structure it acts upon (**digits**), and its depth (**superficial**).

Bones Moved by Back Muscles

- **Axial Skeleton**
  - Vertebral Column

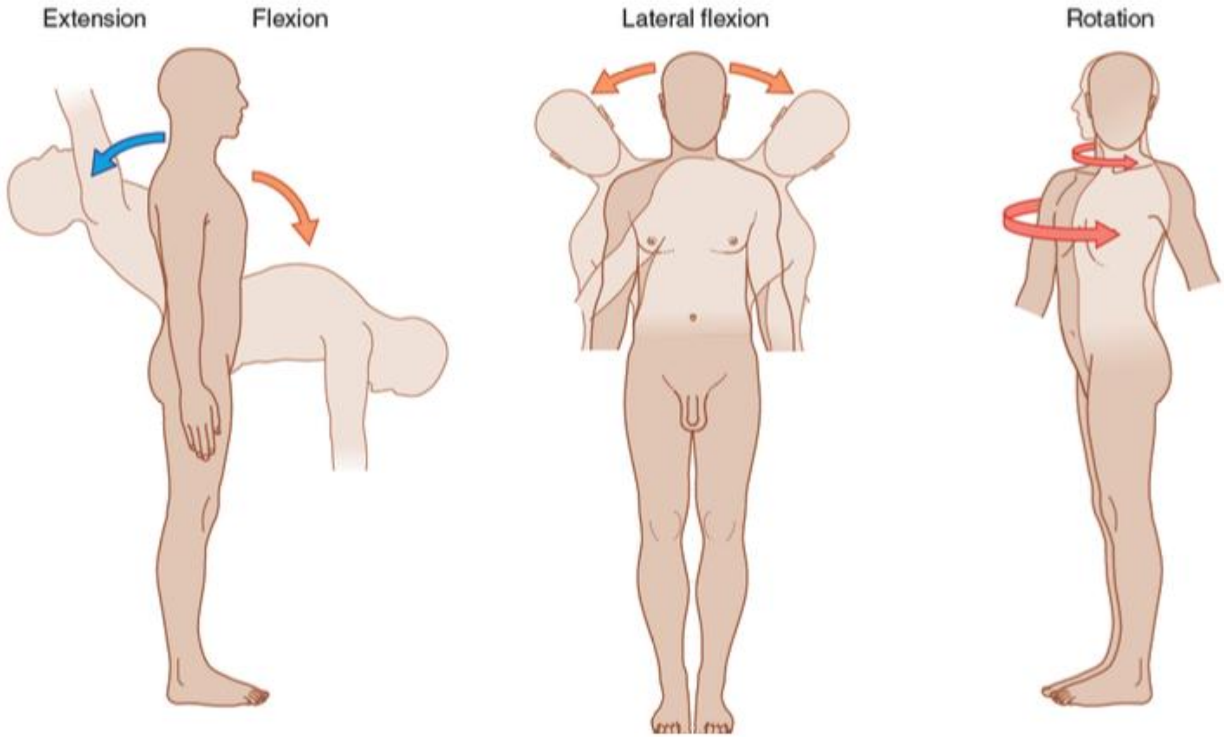
- **Appendicular Skeleton**
  - Pectoral Girdle: Clavicle and Scapula
    - *Acromioclavicular Joint*
    - *Sternoclavicular Joint*
    - *Scapulothoracic Joint*
  - Humerus
    - *Glenohumeral Joint*





Movements of the Vertebral Column

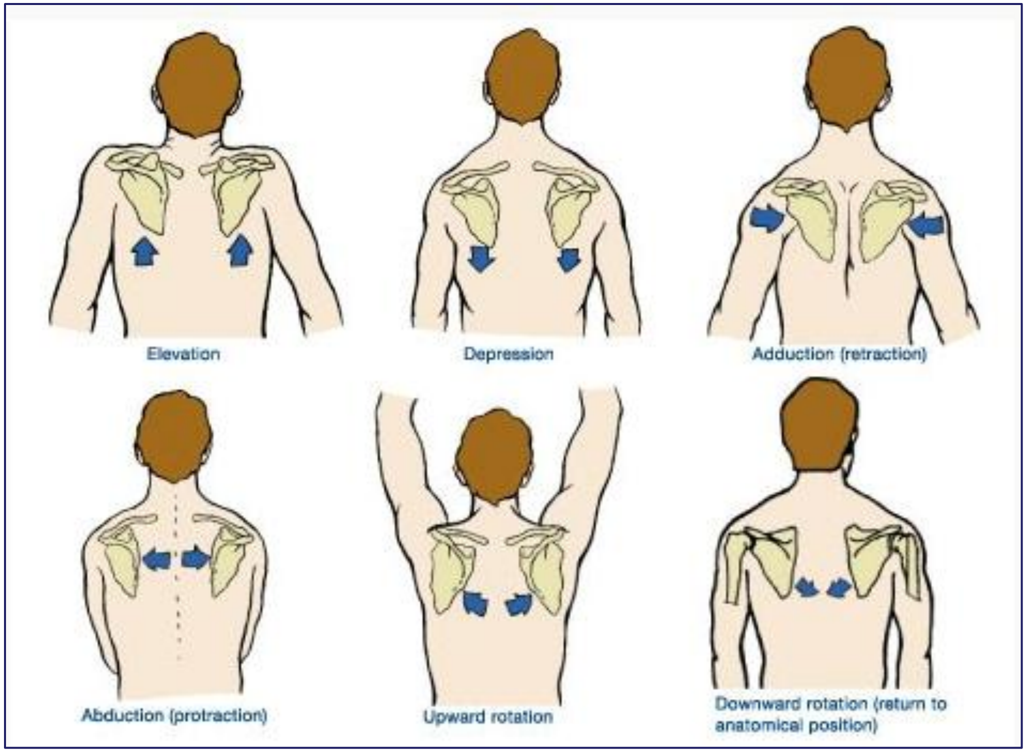
- **Extension** – backward bending
- **Flexion** – forward bending
- **Lateral Flexion** – side bending
- **Rotation** – twisting side to side



Movements of the Shoulder Girdle

Scapula

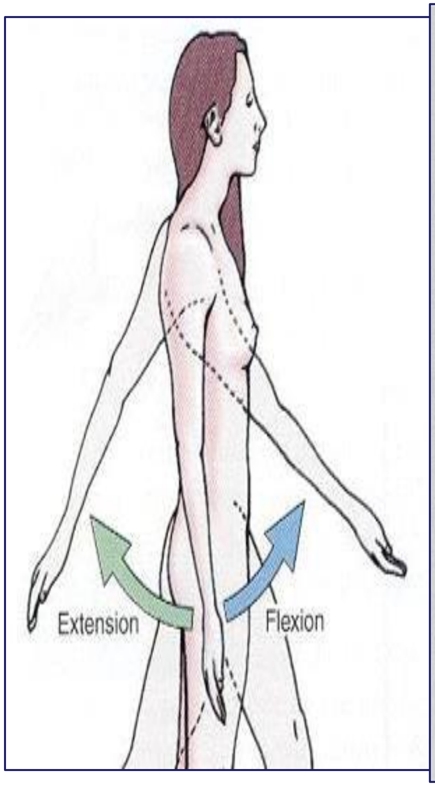
- **Elevation** – cranial movement of scapula
- **Depression** – caudal movement of scapula
- **Adduction/Retraction** – medial borer drawn closer together
- **Abduction/Protraction** – medial border drawn away from each other
- **Upward Rotation** – inferior angle moves laterally and upwards
- **Downward Rotation** – inferior angle moves medially and downwards

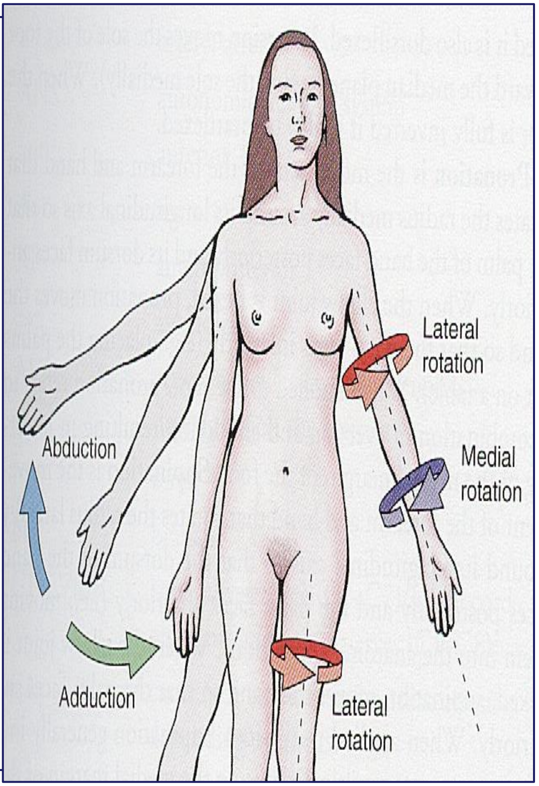


Movements of the Shoulder Girdle

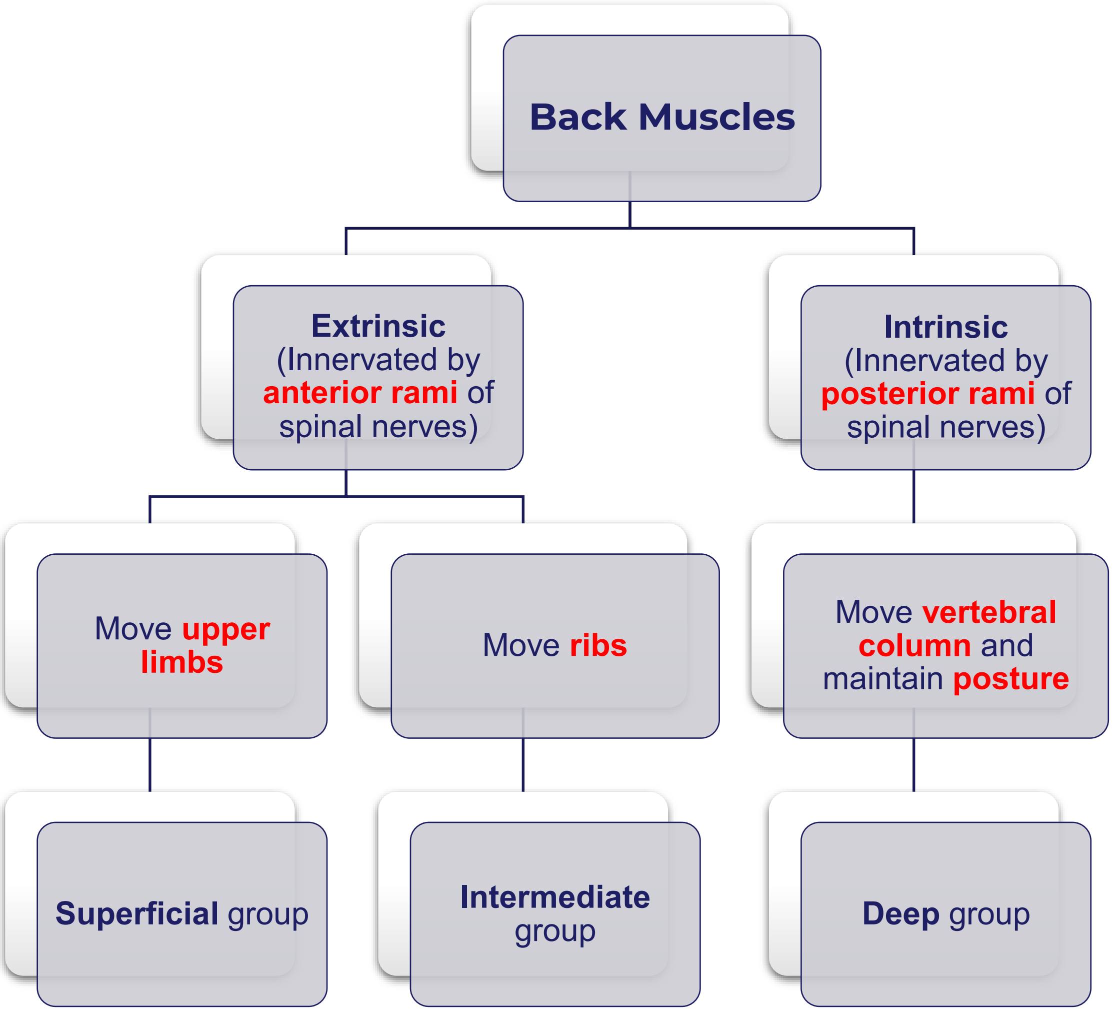
Humerus at the Glenohumeral Joint

- **Extension** – the movement of the humerus straight posteriorly
- **Flexion** – the movement of the humerus straight anteriorly
- **Abduction** – upward lateral movement of humerus out to the side away from the body
- **Adduction** – downward movement of humerus medially toward the body from abduction
- **Lateral (External) Rotation** – movement of humerus around long axis of the bone away from the midline
- **Medial (Internal) Rotation** – movement of humerus around long axis of the bone towards from the midline





Overview of Back Muscles



Overview of Back Muscles

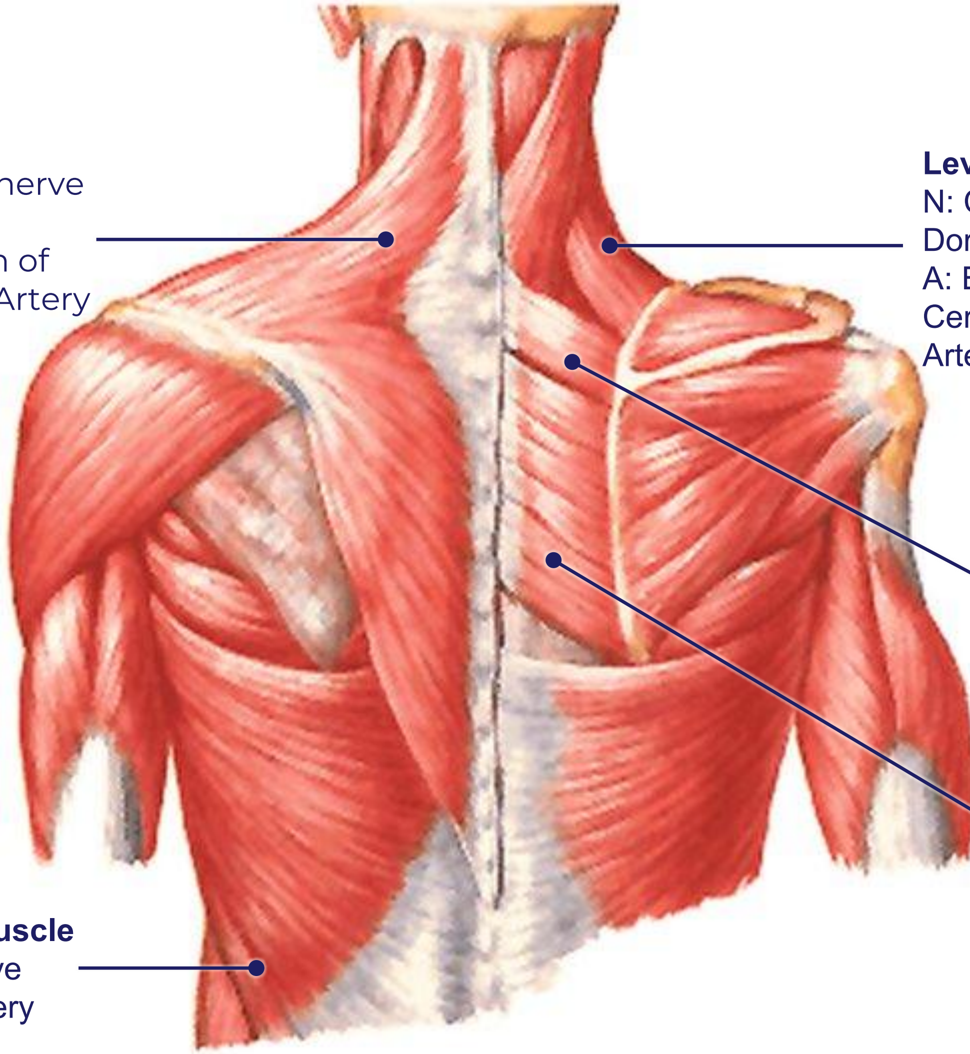
- **Back Muscles**
  - **Extrinsic Group**
    - **Superficial (Appendicular) group**
      - ***1st Layer: Trapezius & Latissimus Dorsi***
      - ***2nd Layer: Rhomboids and Levator Scapulae***
    - **Intermediate (Respiratory) group**
      - ***Serratus Post Sup.***
      - ***Serratus Post Inf.***
  - **Intrinsic Group**
    - **Deep group**
      - **Head and neck** ***(Splenius)***
      - **Vertebral column (*Erector Spinae* & *Transverso spinales*)**
      - **Short segmental muscles** ***(Interspinales & Intertransversarri)***

Extrinsic Back Muscles

- Positioned above or superficial to the thoracolumbar fascia
- Neurovascular Supply
  - Innervated by named anterior (**VENTRAL**) rami branches of the brachial plexus
  - Receive blood supply from branches of the subclavian and axillary artery
- Contains two main muscle groups
  - 1. Superficial (Appendicular) Group
  - 2. Intermediate (Respiratory) Group

Superficial Appendicular Muscles

- These muscles connect the axial skeleton to the appendicular skeleton
- Primarily moves the upper limb by moving the scapula or the humerus
- Arranged into two layers
- Layer 1: *Trapezius Muscle* and *Latissimus Dorsi*
- Layer 2: *Levator Scapulae, Rhomboid Major and Minor*

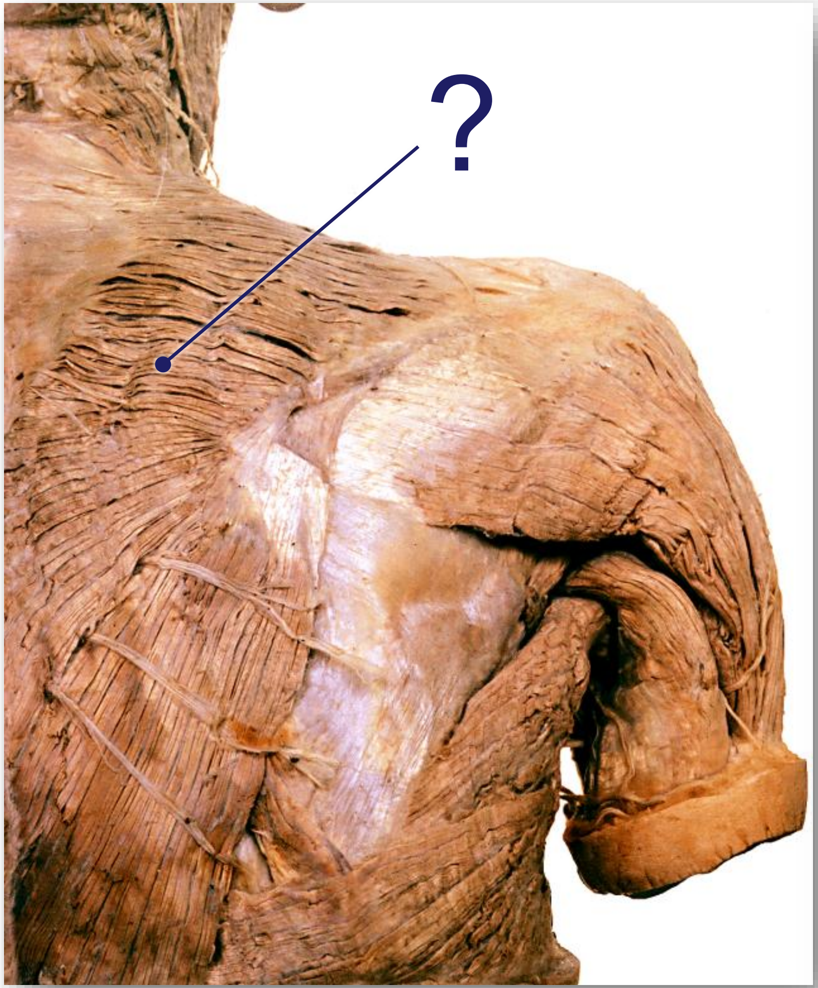

Trapezius Muscle
N: Spinal Accessory nerve (Cranial Nerve 11)
A: Superficial Branch of Transverse Cervical Artery

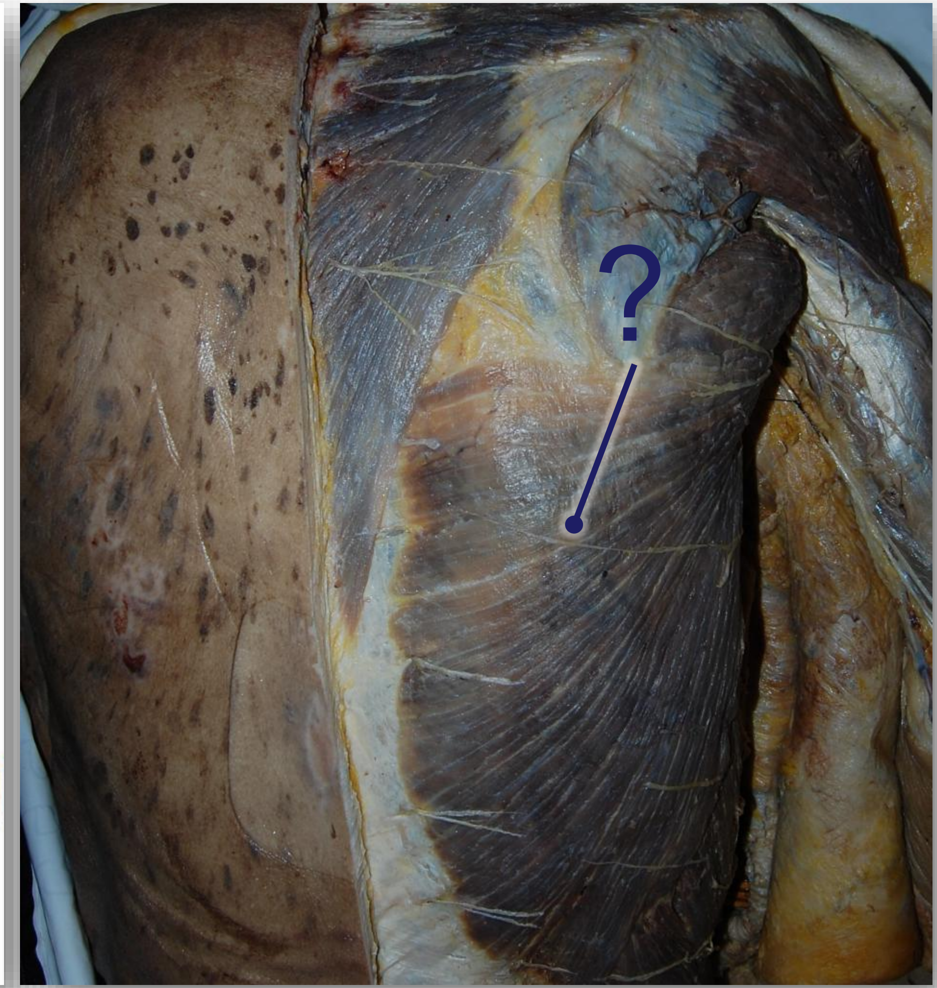
Levator Scapulae
N: C3, C4 Spinal Nerves **AND** Dorsal Scapular Neve
A: Branches of Transverse Cervical and Ascending Cervical Artery

Rhomboid Minor Muscle
N: Dorsal Scapular Neve
A: Deep Branch of Transverse Cervical or Dorsal Scapular Artery

Rhomboid Major Muscle

Latissimus Dorsi Muscle
N: Thoracodorsal neve
A: Thoracodorsal Artery





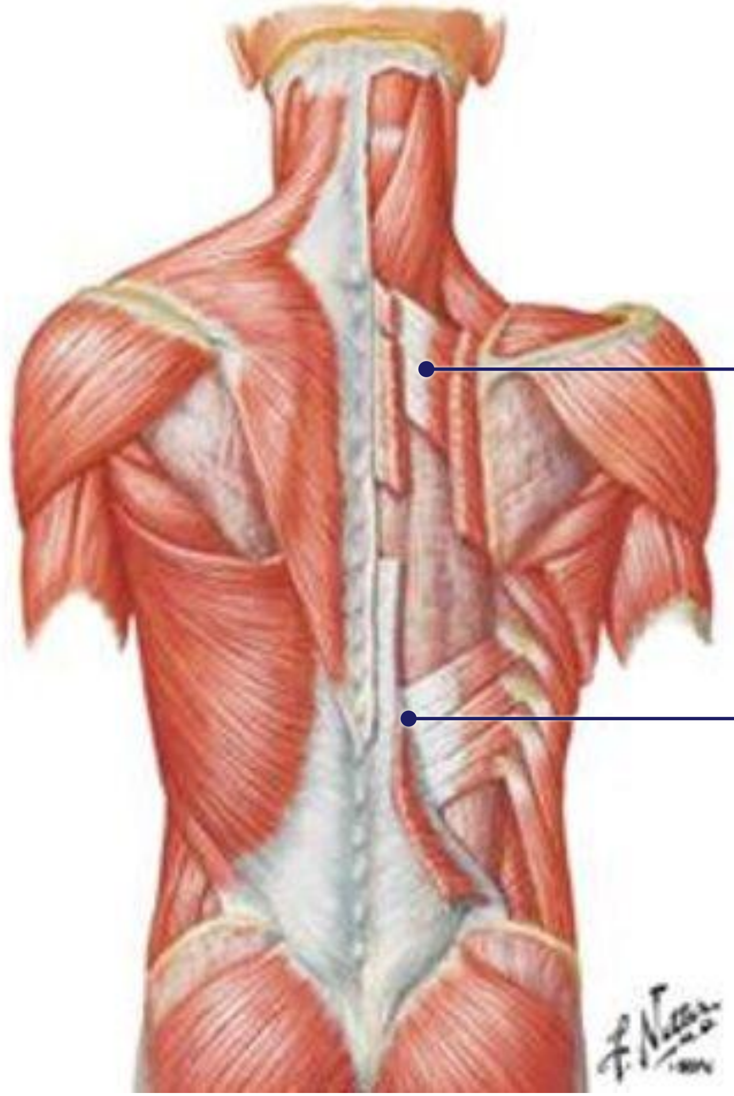
Respiratory or Intermediate Extrinsic Group

- Consists of a thin layer of muscles positioned deep to the superficial appendicular group muscles but is still superficial to the thoracolumbar fascia

- Pass obliquely outward from the vertebral column to ribs.
- Elevate and depress the ribs.
- They DO NOT move the humerus or scapula

- Serve a proprioceptive (sensory) function
- *Proprioception – a subconscious awareness of moments and position of the body independent of visual cues*

- Two muscles make up the respiratory group:
- *Serratus Posterior Superior* – located in the upper back
- *Serratus Posterior Inferior* – located in the lower back



Serratus Posterior Superior

N: Ventral Rami of Intercostal nerves
A: Segmental arterial supply from intercostal arteries

Serratus Posterior Inferior

Summary Table Extrinsic Back Muscles

MUSCLE	FUNCTION	INNERVATION	BLOOD SUPPLY
Trapezius	Assists in rotating the scapula during abduction of humerus above horizontal; •upper fibers elevate the scapula •middle fibers adduct the scapula •lower fibers depress the scapula	•Accessory nerve (motor) •Cervical spinal nerves C3 - C4 (proprioception)	Superficial branch of transverse cervical artery
Latissimus Dorsi	Extends, adducts, and medially rotates humerus	Thoracodorsal nerve (C6-C8)	Thoracodorsal artery
Levator Scapulae	Elevates scapula	•C3 - C4 Spinal Nerves •Dorsal scapular nerve	Transverse and ascending cervical arteries
Rhomboid Major	Retracts (adducts) and elevates scapula	Dorsal scapular nerve (C4-C5)	Deep branch of transverse cervical artery and/or dorsal scapular artery
Rhomboid Minor	Retracts (adducts) and elevates scapula	Dorsal scapular nerve (C4-C5)	Deep branch of transverse cervical artery and/or dorsal scapular artery
Serratus Posterior Superior	Elevates ribs II to V	Anterior rami of upper thoracic (intercostal) nerves T2 to T5	Segmental supply through intercostal art eries
Serratus Posterior Inferior	Depresses ribs IX to XII and may prevent lower ribs from elevating when diaphragm contracts	Anterior rami of lower thoracic (intercostal) nerves T9 to T12	Segmental supply through intercostal art eries

Intrinsic Back Muscles

- Rest deep to the thoracolumbar fascia

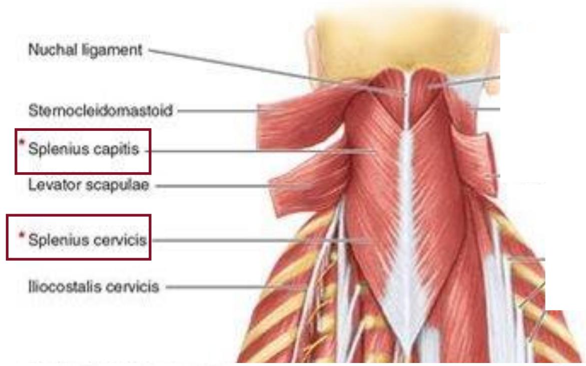
- Involved in movement of "true" back bones (i.e., the vertebral column), and therefore called intrinsic back muscles. Also involved in maintaining the upright posture

- Neurovascular supply
- Innervated by DORSAL rami of spinal nerves
- Receives segmental blood supply from branches of the aorta and vertebral arteries

- Arranged into three groups
    1. Extensors and rotators of the head and neck
    2. Extensors and rotators of the vertebral column
    3. Short segmental muscles

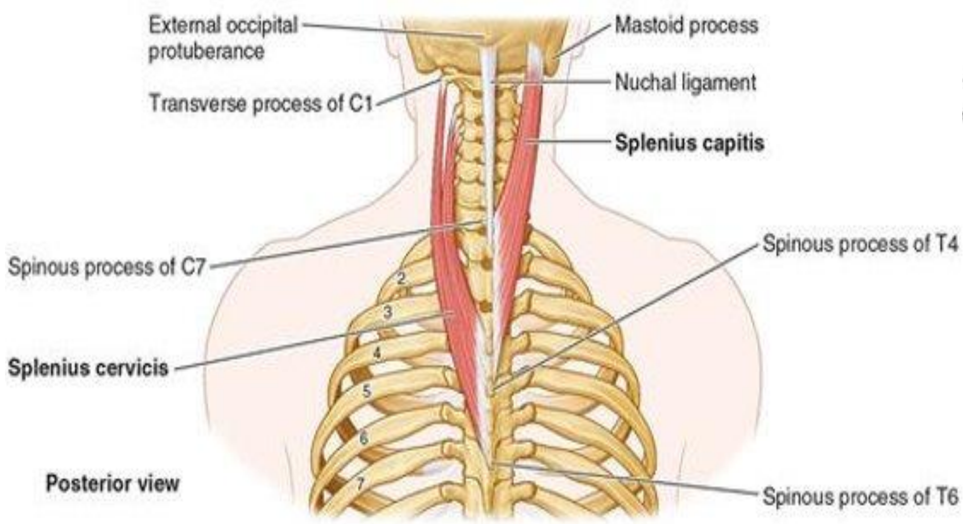
Extensors and rotators of the head and neck

- Splenius Capitis – attaches to bones of the skull
- Splenius Cervicis – attaches to bones associated with the neck



- Bilateral contraction leads to extension of the head, cervical, and upper thoracic vertebrae

- Unilateral contraction causes ipsilateral lateral flexion and rotation (turn face to same side)



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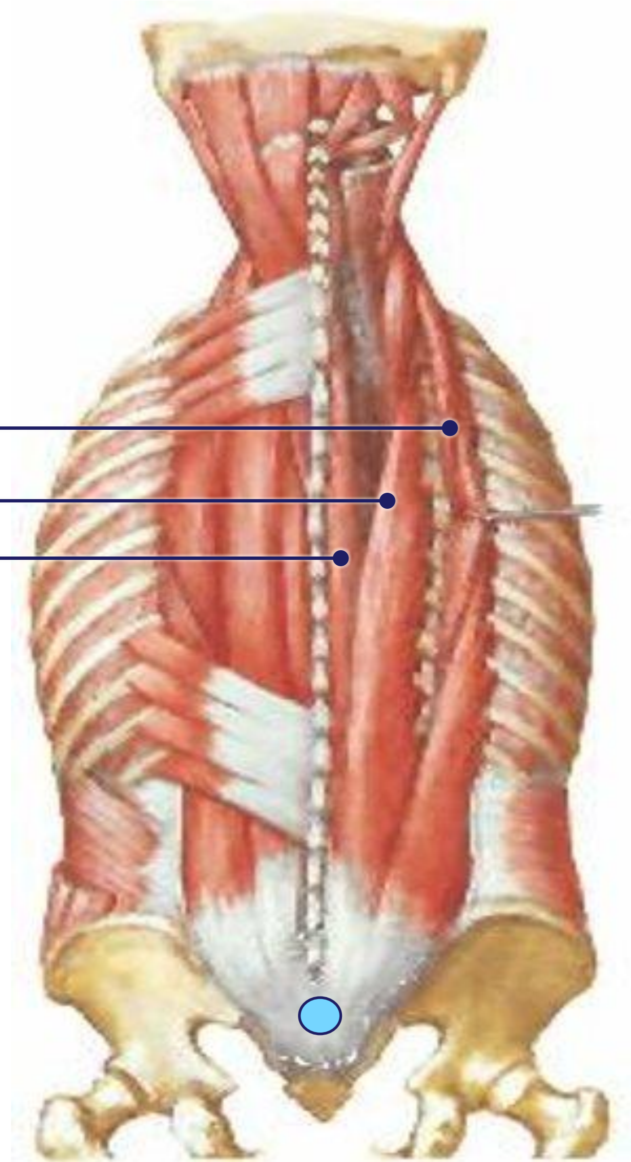
Extensors and rotators of the vertebral column

- Contains 2 layers of muscles:

1. Erector spinae
2. Transversospinales

1. Erector Spinae Muscle (Paraspinal or Paravertebral muscles): main extensors of the vertebral column and are divided into three columns:

- Iliocostalis Muscle (*Lumborum, Thoracic, Cervicis*)
- Longissimus Muscle (*Thoracic, Cervicis, Capitis*)
- Spinalis Muscle (*Thoracis, Cervicis, Capitis*)



- Common origin, a broad tendon that attaches to the iliac crest, sacrum, and inferior lumbar spinous processes

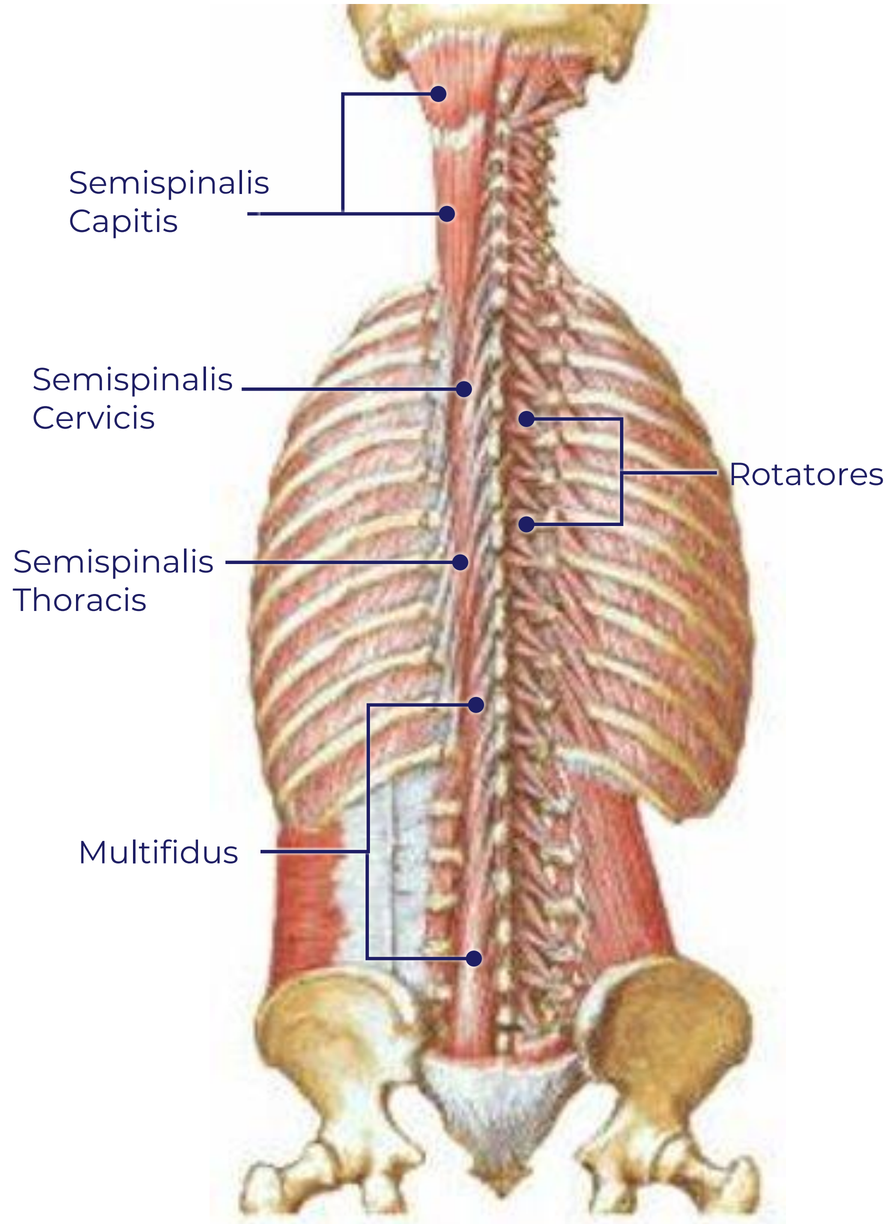
- Bilateral contraction extends / straightens the flexed trunk

- Unilateral contraction causes ipsilateral lateral flexion

Extensors and rotators of the vertebral column

2. Transversospinalis Group

- Found deep to the Erector Spinae Muscles
- Consists of three obliquely orientated muscles
- *Semispinalis (Capitis, Cervicis, Thoracis)*
- *Multifidus*
- *Rotatores (Cervicis, thoracis, lumborum)*
- Semispinalis bilaterally extends the head and vertebral column. Unilateral contraction causes contralateral rotation
- Multifidus and Rotatores provides stability to the vertebral column, assist in local extension of the vertebral column, and unilateral contraction causes contralateral rotation

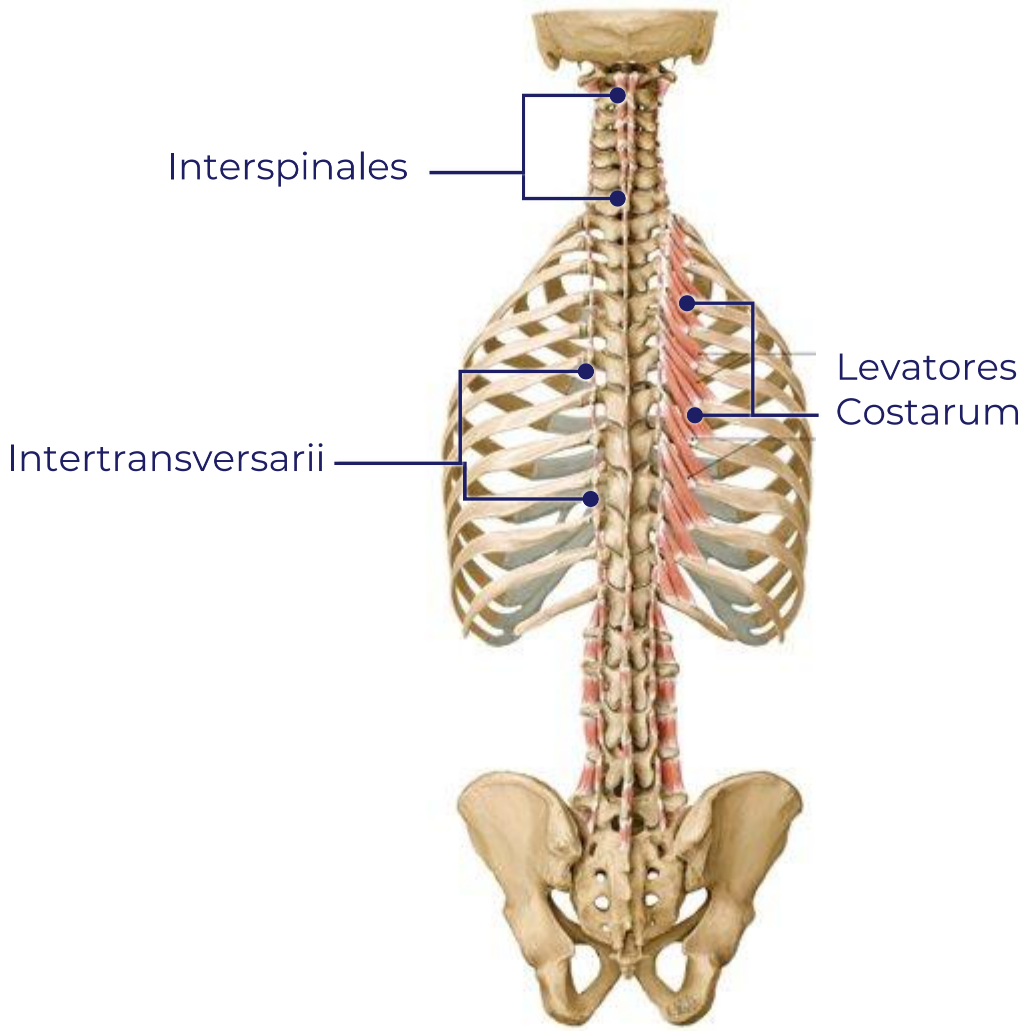


Short Segmental muscles

- Consists of three muscles
  - *Interspinales*
  - *Intertransversarii*
  - *Levator Costarum*

- Interspinales and Intertransversarii – contains many muscle spindles (sensory apparatus) imparting a proprioceptive role
- These muscles are postural muscles that stabilize the vertebrae

- Levator Costarum - helps elevate ribs (respiratory function)



Summary Table Intrinsic Back Muscles

MUSCLE	FUNCTION	INNERVATION	BLOOD SUPPLY (Segmental)
Splenius Muscle	Bilateral: extension of neck Unilateral: ipsilateral lateral flexion and rotation of vertebral column	Posterior (Dorsal) Rami of Cervical Spinal Nerves	Deep Cervical Artery Occipital Artery
Erector Spinae Muscle Group	Bilateral: extension vertebral column and head; during flexion of back allow for control movement by gradually lengthening fibers Unilateral: ipsilateral flexion of vertebral column	Posterior (Dorsal) Rami of Spinal Nerves	Vertebral Artery Deep Cervical Artery Occipital Artery Transverse Cervical Artery Posterior Intercostal Artery Subcostal Artery Lumbar Artery Lateral Sacral Artery

Summary Table Intrinsic Back Muscles

MUSCLE	FUNCTION	INNERVATION	BLOOD SUPPLY (Segmental)
Semispinalis Muscle	Bilateral: Extends head, cervical and thoracic regions of vertebral column Unilateral: Contralateral rotation of the vertebral column contralaterally	Posterior (Dorsal) Rami of Spinal nerves	Deep Cervical Artery Posterior Intercostal Artery Subcostal Artery Lumbar Artery
Multifidus	Stabilizes vertebrae	Posterior (Doral) Rami of Spinal nerves	
Rotatores	Stabilizes Vertebrae and assist with local extension and rotatory movement of vertebral column	Posterior (Dorsal) Rami of Spinal Nerves	
Interspinales Muscle	Postural and stabilizes vertebrae	Posterior (Dorsal) Rami of Spinal Nerves	
Intertransversarii Muscle	Postural and stabilizes vertebrae	Posterior (Dorsal) Rami of Spinal Nerves	
Levatores costarum	Elevation of ribs (respiratory function Assist with lateral flexion of vertebral column	Posterior (Dorsal) Rami of Spinal Nerves	

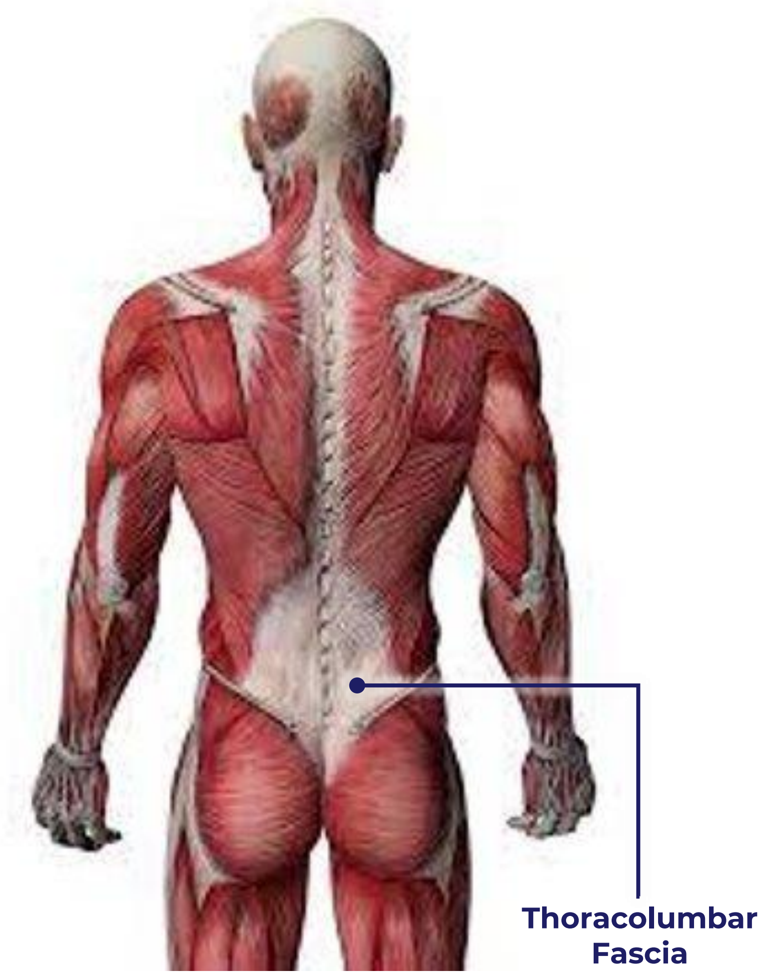
Importance of Fascia

- Interconnects all structures of the body, contains collagen fibers arranged in sheets oriented in different directions
- Functions of fascia (MSK system):
- Allows for the sliding of the muscular structure, and neurovascular structures between contractile fields and joints
- Reduce friction of muscular force
- Forms muscle compartments which increase the force of contraction for groups of muscles

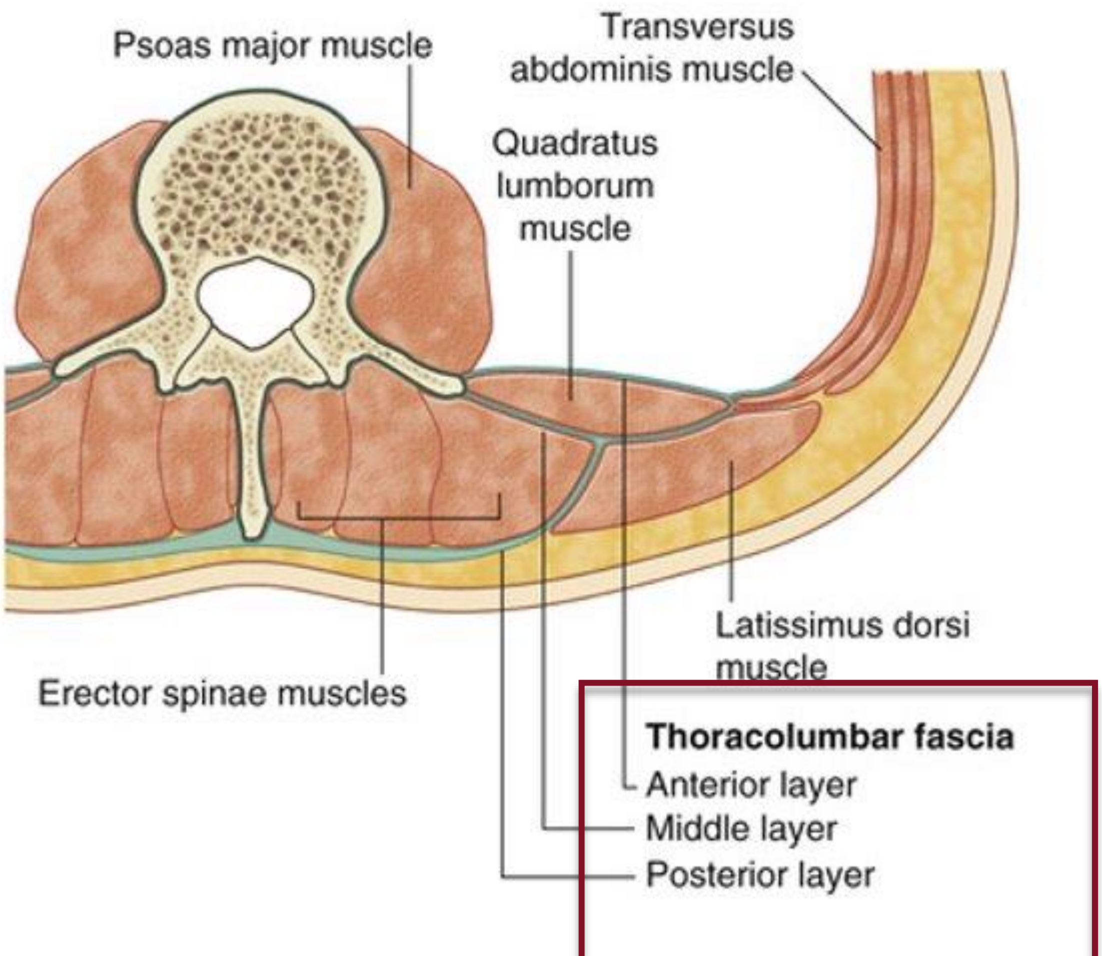
- **Clinical Importance of Fascia**
- Limit the spread of infection and malignant disease
- Injury of fascia lead to the development of scar tissue which can impair muscle contraction
- Fasciotomy – surgical procedure to cut fascia to relieve increasing pressure in MSK compartments

Thoracolumbar Fascia

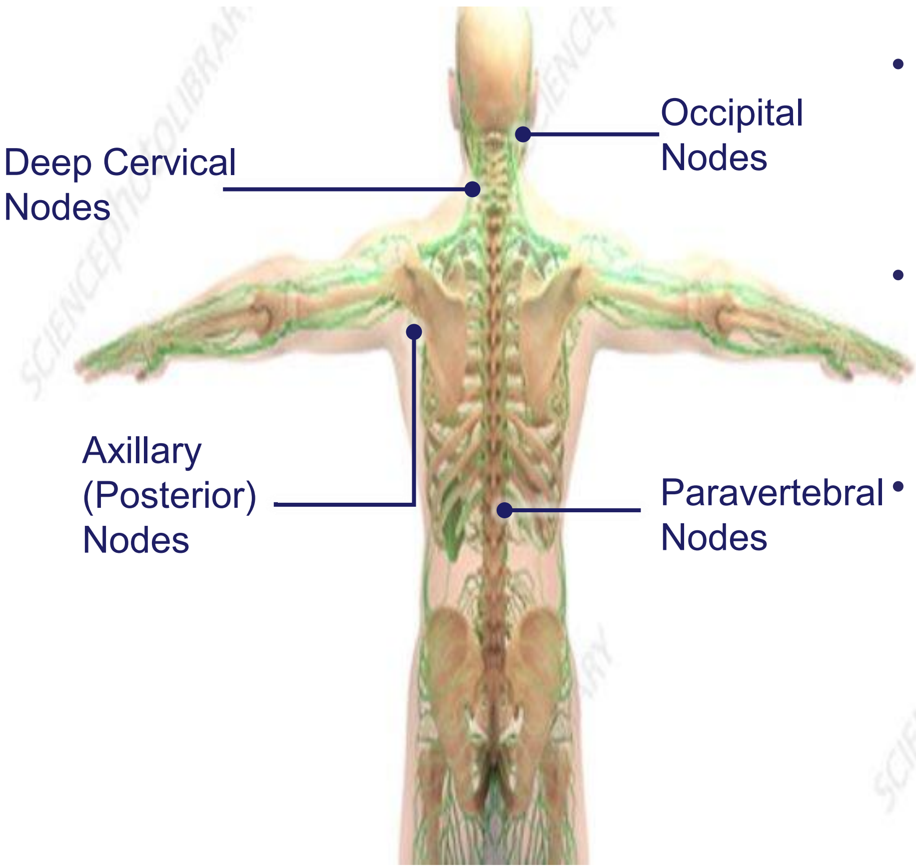
- Deep investing membrane throughout most of the posterior thorax and abdomen
- Formed by longitudinal and transverse fibers from various muscles that bridge the aponeuroses of internal oblique and transversalis muscles. Continuous with deep fascia of the neck (deep cervical fascia)
- Consists of three layers: anterior, middle, and posterior
- The anterior and middle layers insert at the transverse processes of vertebrae
- The posterior layer inserts at the tips of the spinous processes and is indirectly continuous with interspinous ligaments
- Covers or invests the paravertebral muscles (intrinsic muscles) of the back



Thoracolumbar Fascia Layers



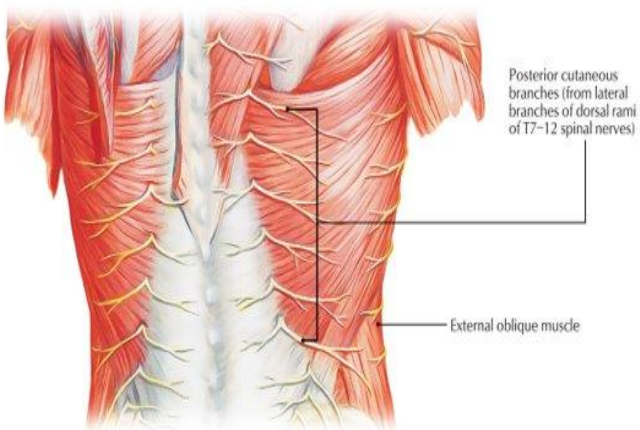
Lymphatic Drainage of Back

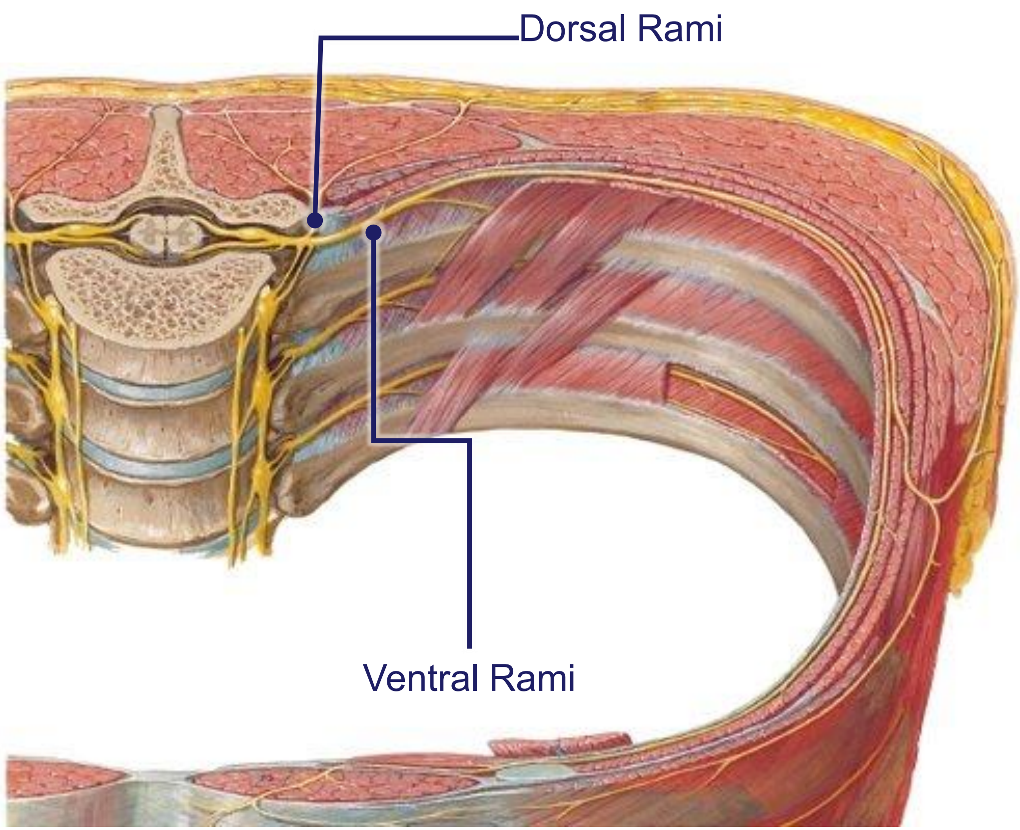


- Lymph from the skin of the back of the neck drains into the occipital, deep cervical, and axillary (posterior) nodes
- Lymph from the skin of the back of the trunk drains into the axillary (posterior) nodes. Some drain into the superficial inguinal nodes
- Lymph from the muscles of the back drain into the paravertebral lymph nodes

Cutaneous Innervation of Back

- The skin of the back region is innervated by the dorsal rami of spinal nerves
- Ventral rami will innervate the lateral body wall via the lateral cutaneous nerve and the anterior body wall via the anterior cutaneous nerve





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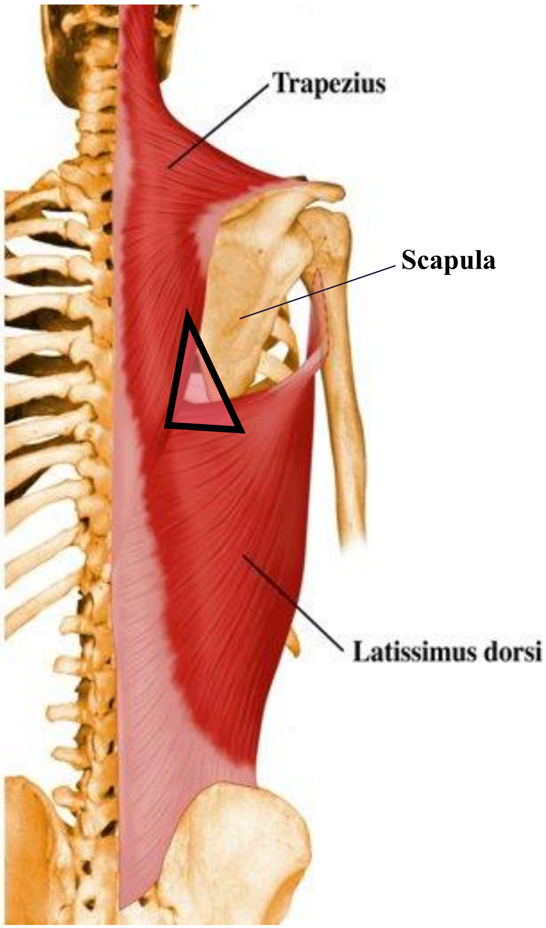
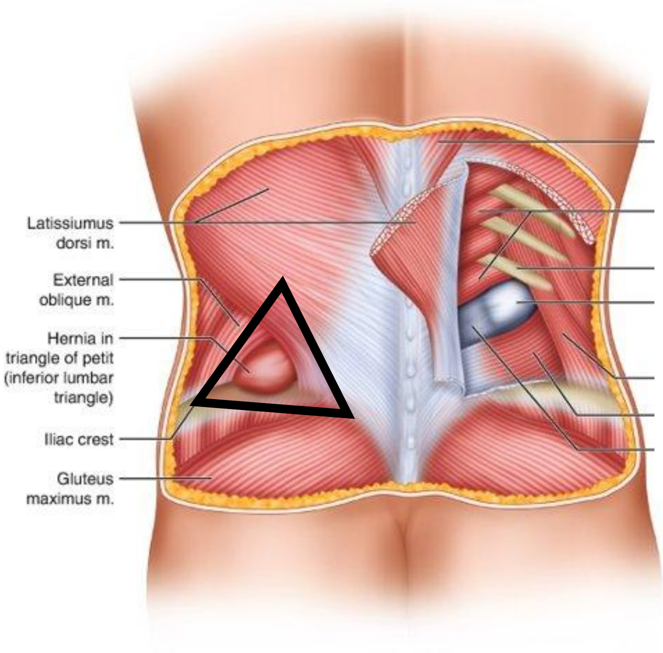
Clinical Anatomy: Triangle of Auscultation



- Anatomical landmark which allows for prime pulmonary auscultation
- Boundaries
- Inferior: Superior boarder Latissimus Dorsi
- Medial: Lateral border of Trapezius
- Lateral: Medial Border of Scapula

Clinical Anatomy: Triangle of Petit

- Also called the Inferior Lumbar Triangle or Lumbar Triangle
- Boundaries
  - Inferior: Iliac Crest
  - Posterior: Latissimus Dorsi muscle
  - Anterior: External Oblique muscle
  - Floor: Internal Oblique Muscle and Transversus Abdominus Muscle
- Petit Hernia (Lumbar Triangle Hernia)
- Appears as a bulge in the lumbar region.
- Can be unilateral or bilateral
- Maybe acquired or congenital
- Rare type of hernia



Surface Anatomy of the Back

A positive mindset brings positive things

Questions Email: mabdelra@sgu.edu